Statistical Intervals Based on a Single Sample (Chapter 5)

In many situations, a point estimate does not provide enough information about the parameter of interest.

An interval estimate can be more informative regarding to the precision and reliability of estimation.

1

Example: IQ test scores and classification

Normal is 100.

Assume that you have a sibling, say X.

X got a score of 98.

You got a score of 101.

Here are some questions:

- Are you the smart guy with superior intelligence?
- Is X dull?

Variability?

IQ Classification
Genius
Very superior intelligence
Superior intelligence
Normal intelligence
Dullness
Border-line deficiency
Definite feeble-mindedness

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Example: IQ test

Average is 100.

Assume that you have a sibling, say X.

X got a score of 98.

You got a score of 101.

Here are some questions:

- Are you the smart guy?
- Is X backward or dull?

Stanford–Binet Fifth Edition (SB5) classification	
IQ Range ("deviation IQ")	IQ Classification
145–160	Very gifted or highly advanced
130–144	Gifted or very advanced
120–129	Superior
110–119	High average
90–109	Average
80–89	Low average
70–79	Borderline impaired or delayed
55–69	Mildly impaired or delayed
40–54	Moderately impaired or delayed

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Confidence Intervals…

In many situations, a point estimate does not provide enough information about the parameter of interest.

An interval estimate can be more informative regarding to the precision and reliability of estimation.

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Remember:

The point estimates for p, where p is the probability of non-defective.



Now the question is:
Can we find an interval that will include the true value of p with some probability?

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Normal random sample

$X_1, \ldots, X_n$: random sample

$X_1, \ldots, X_n$ ~ Normal with mean μ and standard deviation σ.

Assume that σ is known.

You want to find an interval that includes the mean μ with probability 1- α.

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An interval estimator for μ for $X_i \sim N(\mu, \sigma)$

Our goal is to find L and U such that:

$P(L < \mu < U) = 1 - \alpha$, where L, U are random variables.

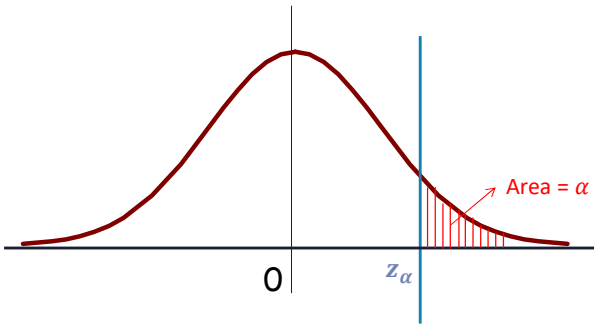
What do we know?

1. An estimator for μ: $\bar{X}$
2. Distribution of the sample mean: $\bar{X} \sim N(\mu, \frac{\sigma}{\sqrt{n}})$

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Definition

z_α : the upper 100α percentage point of the standard normal distribution

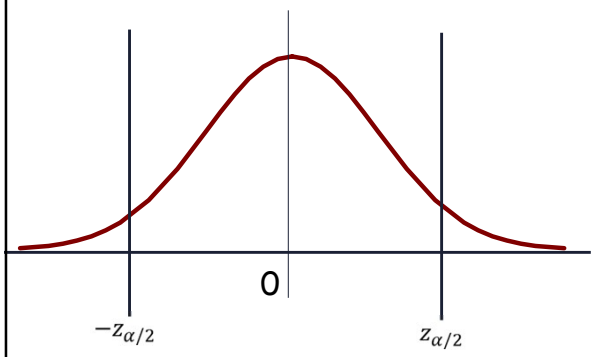


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Question:

$Z \sim N(0,1)$

$P(-z_{\alpha/2} < Z < z_{\alpha/2})$=?



Z in terms of $\bar{X}$?

$\bar{X} \sim N(\mu, \frac{\sigma}{\sqrt{n}})$

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An interval estimator for μ for X_i ~N(μ, σ)

$$P(\bar{X} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}} < \mu < \bar{X} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}) = 1 - \alpha.$$

L U

probability

In practice, only one random sample is used to construct one confidence interval.

$$P(\bar{x} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}} < \mu < \bar{x} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}) = 1 - \alpha.$$

$L=l$ $U=u$

confidence

All quantities in this expression are constants, no random variable is involved.

→ NOT REASONABLE TO ATTACH A PROBABILTY TO THIS SPECIFIC EVENT.

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Definition

A 100(1- α)% CI for the mean μ of a normal population when σ is known is given by:

$$\left(\bar{x} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}}, \bar{x} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right)$$

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Example: A cargo company

A cargo company aims to have a mean delivery time of 5 hours for local deliveries.

They believe that delivery times are normal with σ=2.4.

They recorded 10 delivery times as:

7.3, 3.3, 4.7, 6.8, 10.1, 5.7, 6.4, 4.2, 3.1, 8.9

95%CI on the true mean delivery time?

$$\bar{x} = 6.05$$

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The correct interpretation is by many samples via frequency...

Assume that M samples are collected, each having n observations in it.

We use the above formula to calculate a confidence interval for each of these samples.

Here is a description, and
you will do this in the lab yourself...

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M samples, each with n observations, and 1-α as the confidence level

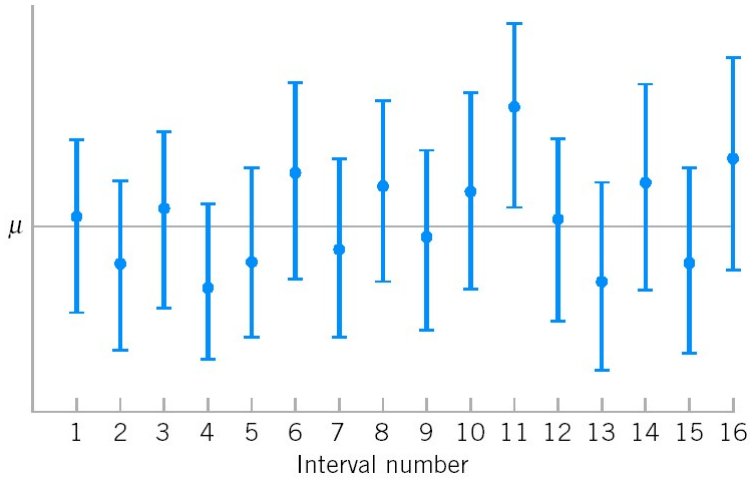
	Observations				Mean	lower bnd	upper bnd
Sample	1	2	...	n	$\bar{x}$	l	u
1	x_1^1	x_2^1	...	x_n^1	$\bar{x}^1$	l^1	u^1
2	x_1^2	x_2^2	...	x_n^2	$\bar{x}^2$	l^2	u^2
M-1	x_1^{M-1}	x_2^{M-1}	...	x_n^{M-1}	$\bar{x}^{M-1}$	l^{M-1}	u^{M-1}
M	x_1^M	x_2^M	...	x_n^M	$\bar{x}^M$	l^M	u^M

How many intervals will cover the real μ?

$\bar{x}^M - z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$ $\quad$ $\bar{x}^M + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$

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Confidence Interval on the Mean of a Normal Distribution, Variance Known



Repeated construction of a confidence interval for μ.

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Confidence Interval on the Mean of a Normal Distribution, Variance Known

Interpreting a Confidence Interval

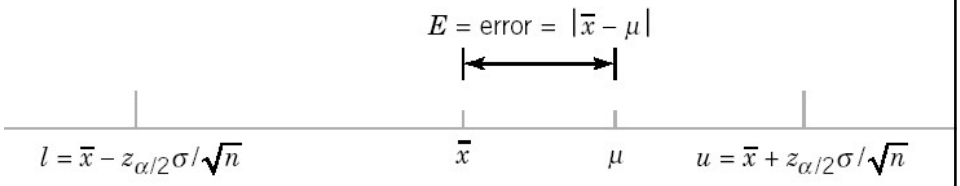
- The confidence interval is a random interval.
- The appropriate interpretation of a confidence interval (for example on μ) is: The observed interval $[l, u]$ brackets the true value of μ, with confidence 100(1-α).

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Confidence Interval on the Mean of a Normal Distribution, Variance Known

Confidence Level and Precision of Error

The length of a confidence interval is a measure of the precision of estimation.



Error in estimating μ with $\bar{x}$.

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$$100(1-\alpha)\%\text{CI:} \quad \left(\bar{x} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}}, \bar{x} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right)$$

Confidence Level & Precision

Confidence Level	Width of the Interval
100(1- α)%	$2z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$
95%	$2(1.96)\frac{\sigma}{\sqrt{n}}$
99%	$2(2.58)\frac{\sigma}{\sqrt{n}}$
100%	∞

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Confidence Level, Precision & Choice of Sample Size

For fixed sample size: If the confidence level is increased, the width of CI is also increased so that the precision or accuracy (measured by the width of the CI) decreases if the reliability of CI is increased.

To have both high confidence & high precision, the only way is to increase the sample size.

$$\left(\bar{x} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}}, \bar{x} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right)$$

Width of the CI: $2z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$

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